

Engineering Chemistry

Unit $\rightarrow$ 4

Spectroscopy

Study of a compound using spectra of the electromagnetic radiation with metals.

→ Electromagnetic Radiation

It's a form of radiant energy which has both particle and wave nature. It's made by the interaction of electric & magnetic fields.

$$(E \perp B)$$



→ Interaction of EM wave and matter

This interaction is of two types

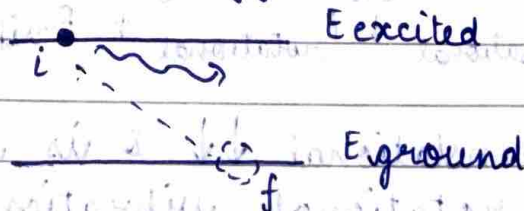
1. Absorption

Electron takes up external energy and becomes excited



2. Emission

Electron releases energy and becomes de-excited



$$E_{\text{excited}} > E_{\text{ground}}$$

→ electromagnetic spectrum

electromagnetic spectrum is obtained when radiations are arranged in order of increasing wavelength or decreasing frequencies

cosmic γ gamma $\times$ Ray γ U.V γ Visible γ Infrared γ Micro Radio
 $f \downarrow \quad \uparrow$

1 radiowave is highest & γ radiowave is lowest

→ Advantages of spectroscopy

1. Important tool to study atomic & molecular structure in complex organic compounds
2. Less time consuming
3. Very small amount of sample is required
4. Cost effective in long run

→ Spectra

The energy change or frequency of EM radiation emitted or absorbed can be recorded with help of instrument called spectrophotometer. This recorded evidence is called spectra.

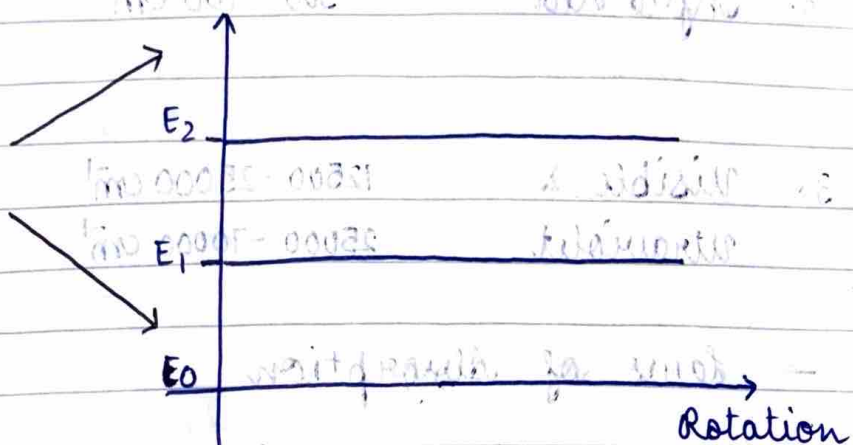
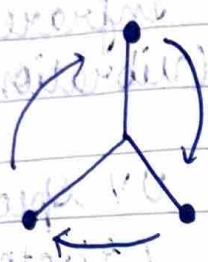
→ Types of energy changes

$$E_{\text{Total}} = E_{\text{translational}} + E_{\text{rotational}} + E_{\text{vibrational}} + E_{\text{electronic}}$$

But $E_{\text{translational}} \downarrow \downarrow$ & is neglected so we only consider rotational, vibrational & electronic making 3 types of spectra:-

1. Pure rotational
2. vibrational rotational
3. Electronic band spectra

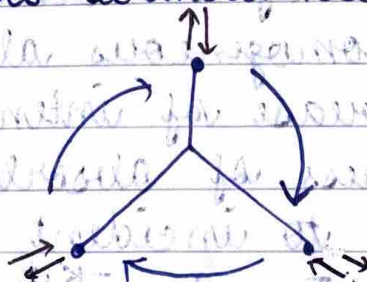
- Rotational Spectra
Spectra generated due to change in rotational energy while keeping others constant is called rotational spectra.



It's observed in far infrared and microwave regions.

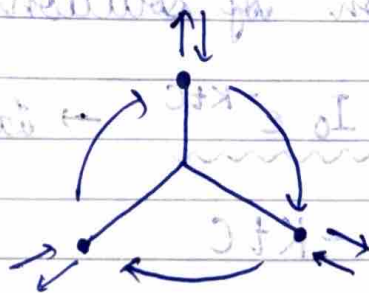
→ I/R spectra

- Vibrational & Rotational Spectra
Generated due to both vibrational & rotational energy changes without electronic transition.



It's observed in near infrared region.

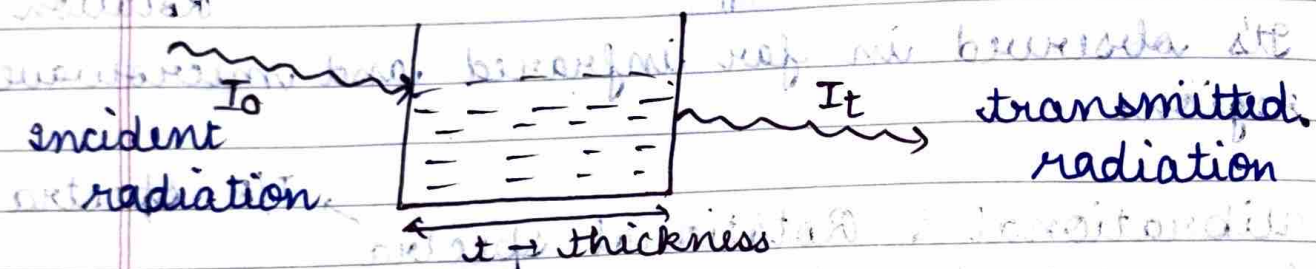
- Translational Spectra (Electronic Band Spectra)
When the changes in energy result in electronic transition.



It's observed in visible and ultraviolet regions.

- Spectral Region
- | | | |
|--------------------------|--|-------------------------------------|
| 1. Microwave | 1-100 cm^{-1} | Microwave Spectroscopy (Rotational) |
| 2. Infra Red | 500-400 cm^{-1} | Infrared (vibrational-Rotational) |
| 3. Visible & ultraviolet | 12500-25000 cm^{-1}
25000-70000 cm^{-1} | UV Spectroscopy (Electronic) |

→ Laws of Absorption



→ When a beam of monochromatic light passes through a homogeneous absorbing medium, the rate of decrease of intensity of radiation with thickness of absorbing medium is proportional to incident radiation.

$$\text{i.e. } I_t = I_0 e^{-Kx}$$

$K \rightarrow$ absorption coeff

→ When a beam of light passes through a solution the decrease of intensity is directly proportional to concentration of solution

$$I_t = I_0 e^{-KtC} \rightarrow \text{Lambert - Beer Law}$$

$$\therefore \ln \frac{I_t}{I_0} = -KtC$$

Also $\frac{I_t}{I_0} = T \rightarrow$ Transmittance

$$\Rightarrow -\ln T = K \cdot l \cdot C$$

$$\Rightarrow -\log T = \frac{K \cdot l \cdot C}{2.303} \quad \left[\frac{K}{2.303} = a \right]$$

$[-\log T = A]$ also Absorptivity
 $\hookrightarrow$ absorbance

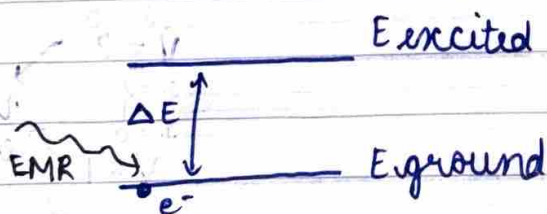
Hence,

$$\underline{A = a \cdot C \cdot l}$$

$\rightarrow$ Principle of spectroscopy

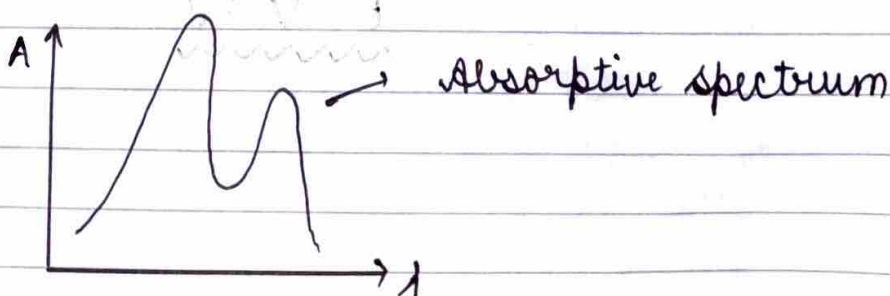
EM radiation provides energy equal to energy difference b/t the energy levels of ground and excited states.

$$\Delta E = h \nu$$



$\rightarrow$ Spectrum

A spectrum is a plot of intensity of absorbance of EM radiation as function of λ over a particular range.



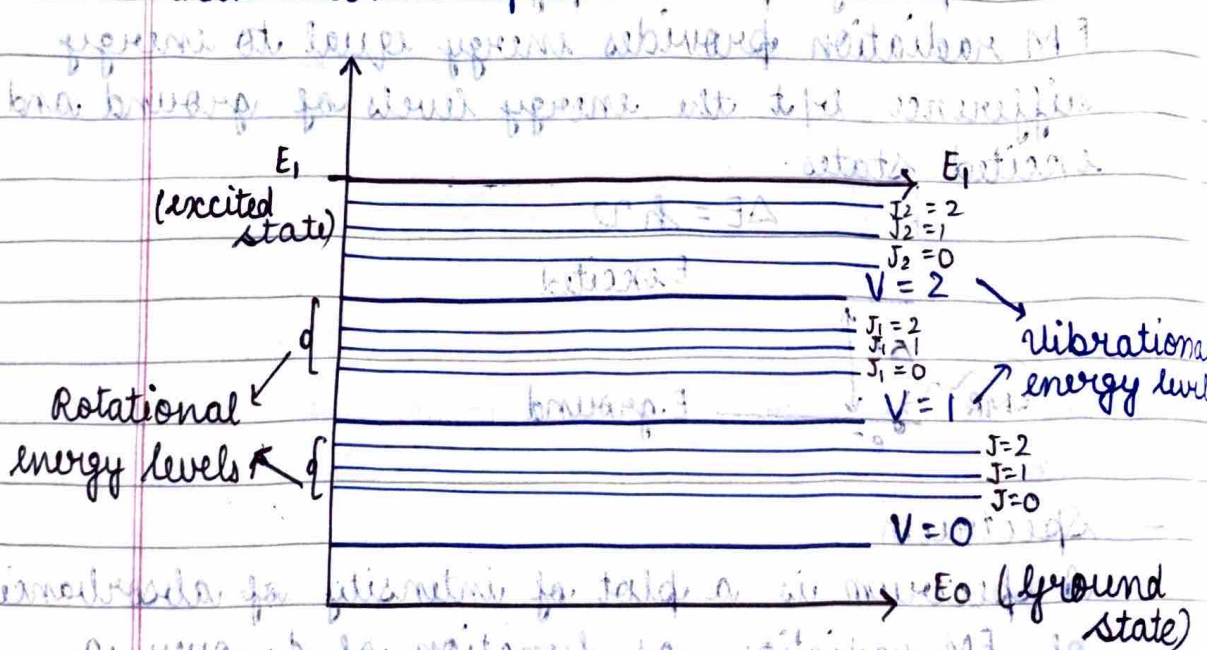
→ Types of spectroscopy

• Atomic spectroscopy

It arises due to transition of electrons between atomic energy levels. It only involves electronic transitions.

• Molecular spectroscopy

It arises due to transition of electrons between molecular energy levels. Energy of molecule can change not only due to e^- transitions but also due to change in rotational and vibrational states. So, it provides more info than atomic spectra.



Also for transition

$$E > V > J$$

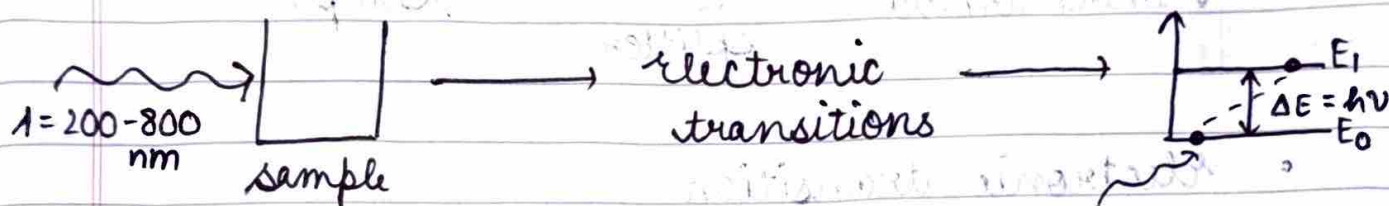
→ U-V visible spectroscopy
 1 UV → 200 - 400 nm
 1 visible → 400 - 800 nm

→ Absorption spectroscopy
 ↳ electronic spectroscopy
 (∵ takes place due to electronic transition)

• Principle

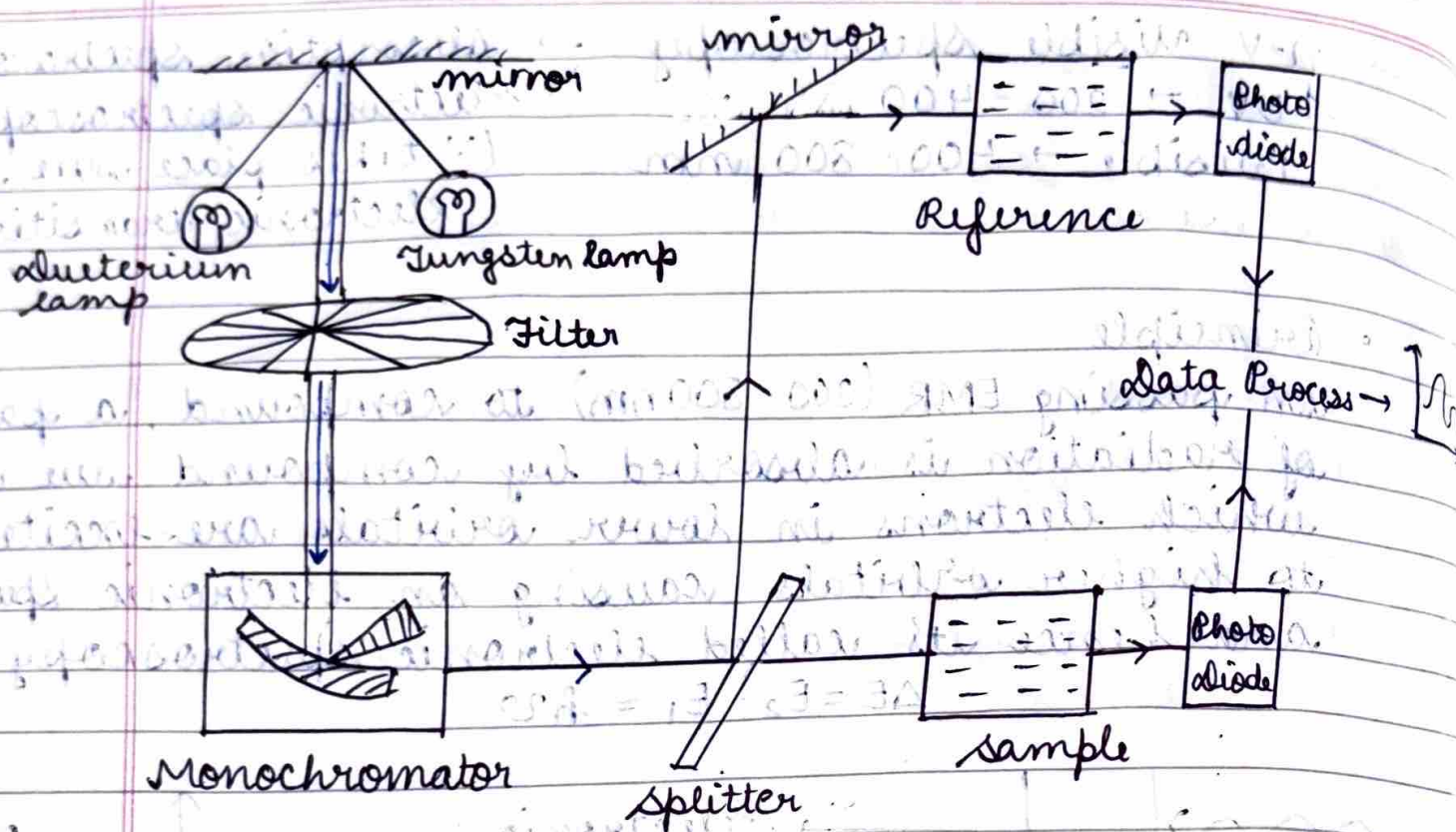
On passing EMR (200-800 nm) to compound, a part of radiation is absorbed by compound due to which electrons in lower orbitals are excited to higher orbitals causing an electronic spectra and hence it's called electronic spectroscopy.

$$\Delta E = E_2 - E_1 = h\nu$$



• Instrumentation

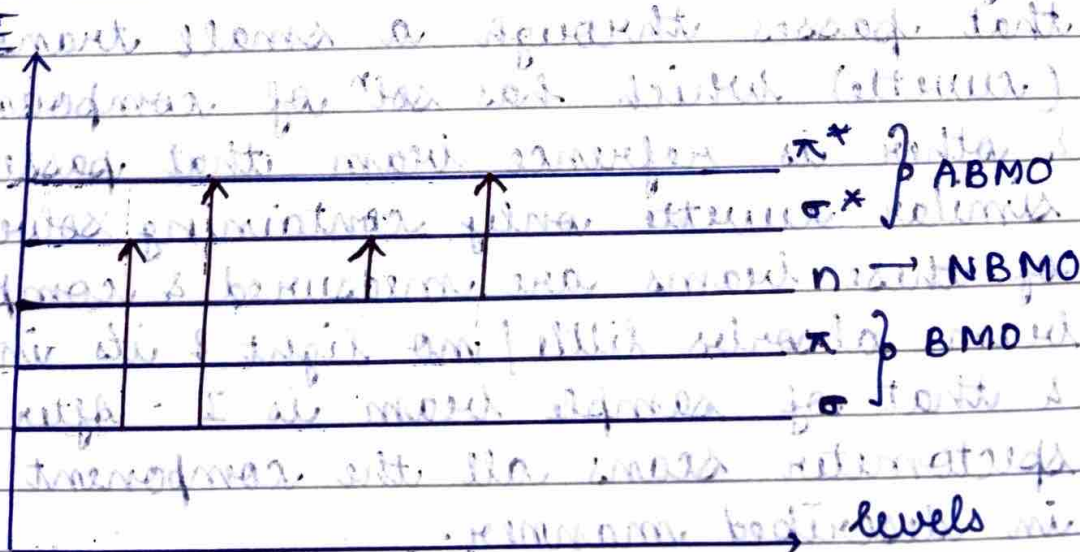
A beam of light is separated into its component wavelengths by a prism and each monochromatic beam is split into two equal intensity beams by half mirrored device; one is the sample beam that passes through a small transparent container (cuvette) which has solⁿ of compound being studied & other is reference beam that passes through a similar cuvette only containing solvent. Intensity of these beams are measured & compared. Reference beam absorbs little/no light & its intensity is I_0 & that of sample beam is I . After some time spectrometer scans all the component wavelengths in described manner.



• Electronic transition

Visible-UV radiation occurs at 200-800 nm and hence gets exposed to molecules; so π present in the bonding & non bonding molecular orbitals enter anti bonding MO like σ^* & π^* and gives rise to following spectra:

$\sigma \rightarrow \sigma^*$ $\pi \rightarrow \pi^*$ $n \rightarrow \pi^*$ $n \rightarrow \sigma^*$



1. $\sigma \rightarrow \sigma^*$ $\Delta E \uparrow$
 due to large energy gap, very high energy is req
 so this transition is seen below 200 nm in
 compounds like saturated hydrocarbons.

2. $n \rightarrow \sigma^*$
 It's seen in saturated molecules with hetero atoms
 like N or O having lone pair electrons. It req
less energy and seen in amines & alcohols.

3. $\pi \rightarrow \pi^*$
 It's seen in unsaturated compounds like alkene
 and alkyne.

4. $n \rightarrow \pi^*$
 An e^- from lone pair of hetero atom is excited
 to π^* . It's seen in ketones.

* $\rightarrow$ Chromophores

- It's a group of molecules responsible for absorption of light by molecule.
- Chromophores exhibit absorption of EM radiation in visible - UV region.
- It may / may not impart colour to the compound.
- They undergo $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ transitions.
- eg $\rightarrow$ ethylene, carbonyl group, Azo compounds.

$\rightarrow$ Auxochromes

- They are the groups which itself don't act as a chromophore, but when it's attached to a chromophore, it shifts the absorption maximum.

towards longer wavelength along with an increase in intensity of absorption. for eg $-OH$, $-NH_2$, $-OR$ groups.

- All auxochromes have one or more pairs of non bonding electrons.

→ * Shifts in UV spectroscopy

Four types of shifts are observed:

1. Bathochromic (Red) Shift

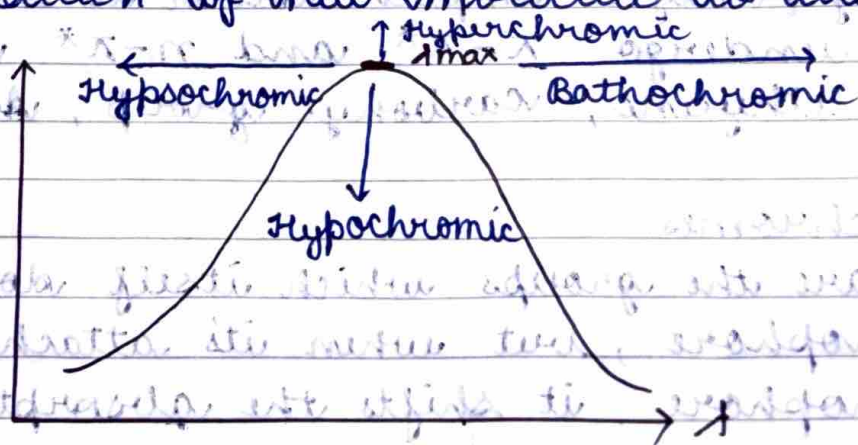
Phenomena which shifts absorption maxima towards longer wavelength due to presence of chromophores, auxochromes or change in solvent eg $\rightarrow n \rightarrow \pi^*$ shift in ketones.

2. Hypsochromic shift (Blue shift)

Phenomena which shifts absorption maxima (λ_{max}) towards shorter wavelength due to removal of conjugation or change in polarity of solvents.

3. Hyperchromic Shifts
Phenomena by which intensity of absorption maxima increases generally due to addition of an auxochrome.

4. Hypochromic shift
Phenomena by which intensity of absorption maxima decreases due to distortion of geometry by addition of new molecule to the group.



→ Applications of UV spectroscopy

- Identification of functional groups
- Identification of unknown compound by taking known compound as reference.
- To detect impurities by showing additional peaks
- To find change in concentration as fun^c of time
- To detect effect of conjugation over λ_{max} .
- To determine amount of calcium in blood serum
- Determination of molar mass.

→ ★ Fluorescence $\rightarrow$ singlet ground $\xleftarrow{\text{radiation}} \leftarrow$ singlet excited $\uparrow$ (visible) $10^{-10} - 10^{-7}$ sec

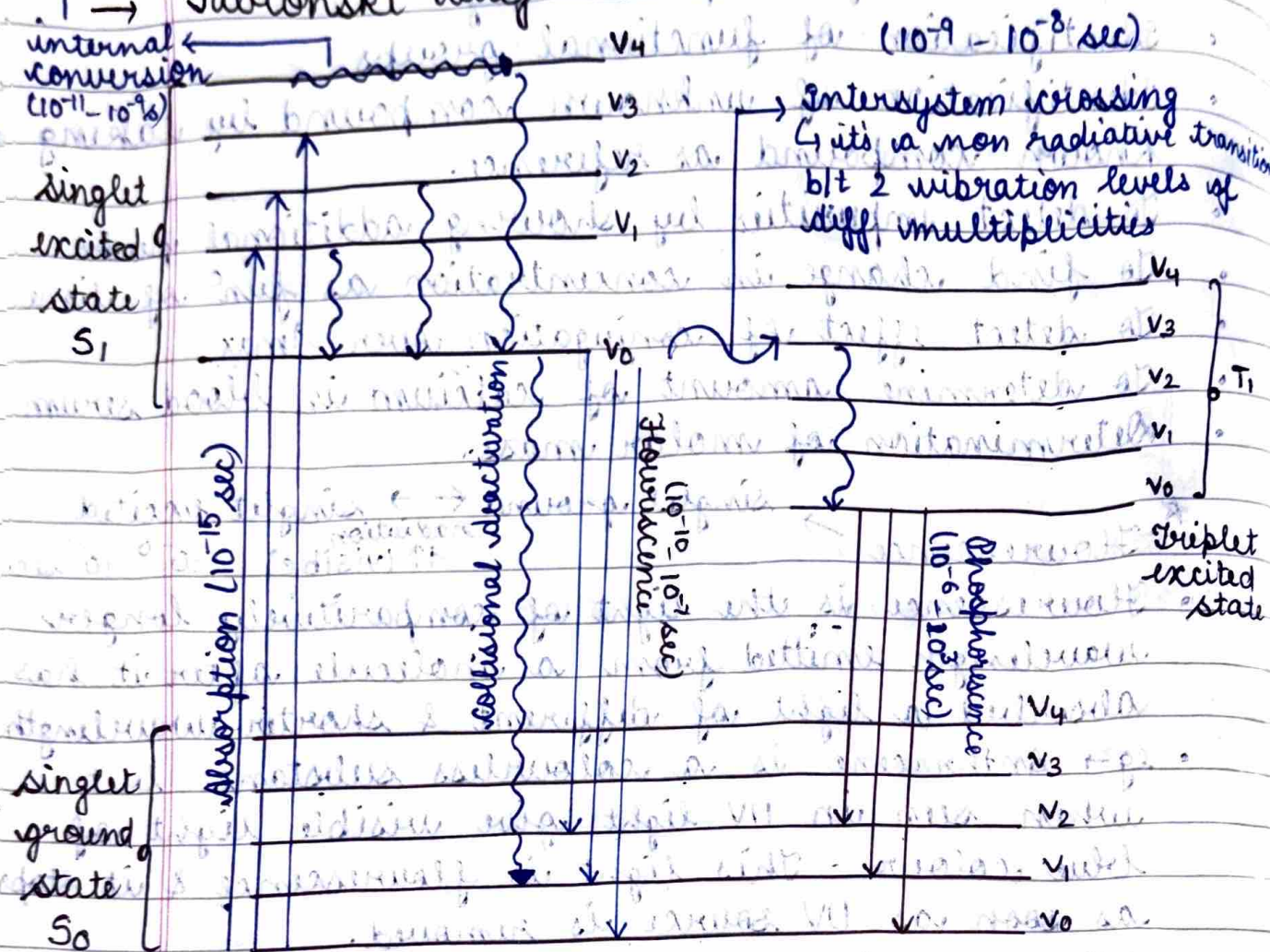
- Fluorescence is the light of comparatively longer wavelength emitted from a molecule after it has absorbed a light of different & shorter wavelength
- eg $\rightarrow$ Anthracene is a colourless substance but when seen in UV light give visible light of blue colour. This light is fluorescence & it stops as soon as UV source is removed.
- When e^- gains $h\nu$ energy and is excited to upper levels, when this e^- returns back to ground state it emits fluorescent radiation of longer wavelength or lower energy than $h\nu$ & causes fluorescence.

→ Applications of fluorescence

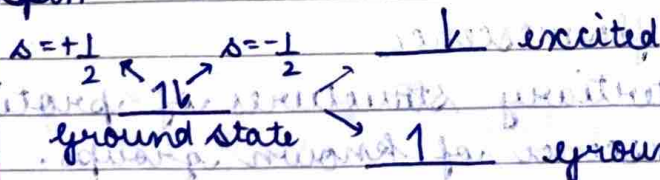
- Study of 3-D tertiary structures of proteins by measuring fluorescence of known groups.
- Early cancer detection & prevents recurrence
- Early detection for tumour removal from tissues.
- In laser treatment
- Angiography & molecular imaging

Transition from excited to
ground state of same
multiplicity

→ Jablonski diagram



→ Singlet state → spin multiplicity = 1
Absence of unpaired electrons
A state in which all unpaired e^- have opposite spin



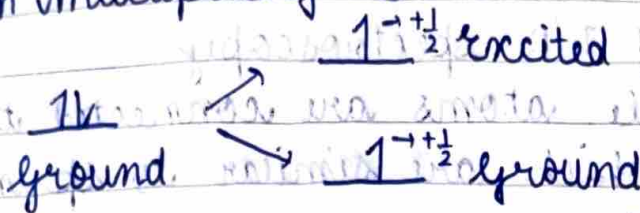
$$s_{\text{net}} = \frac{+1}{2} + \frac{-1}{2} = 0 \text{ (S)}$$

spin multiplicity = $2s + 1$

$$\neq S = 2(0) + 1 = 1 \text{ (singlet)}$$

→ Triplet state

- A state in which unpaired e^- have same spin
- spin multiplicity = 3



$$S_{\text{net}} = \frac{1}{2} + \frac{1}{2} = 1$$

$$\neq S = 2(1) + 1 = 3 \text{ (triplet)}$$

→ Triplet state is more stable than singlet state

★ Phosphorescence

- The emission of radiation when there is transition from triplet excited state to singlet ground state
- energy absorbed is released relatively slow in form of light
- It has avg life time of $10^{-6} - 10^3$ sec

→ Vibrational + Rotational Spectroscopy (Infrared)

- It's a type of absorption spectroscopy which gives a large number of absorption bands using infrared radiations for determination of structure of a compound.
- IR radiation doesn't have enough energy as in the UV radiation so absorption is limited to compounds with small energy difference in possible rotational & vibrational states.
- For a molecule to absorb I/R radiation, the vibration or rotation in molecule must cause a net change in dipole moment of molecule.

1. I/R active $\rightarrow$ changes dipole moment (CO_2 , H_2O , HCl)
2. I/R inactive $\rightarrow$ no change in μ (O_2 , N_2 , Cl_2)

$\rightarrow$ Principle of I/R spectroscopy

In a molecule, atoms are connected to each other by bonds which are similar to springs & not rigid in nature



Due to continuous motion of molecule, the atoms maintain some vibrations with frequency to every portion of molecule. This is called natural freq. of vibration. When energy in form of I/R radiation is applied, it causes vibration when

Applied I/R freq. = Natural freq. of vibration then absorption happens & peak is observed

$\rightarrow$ I/R region of EM spectrum

0.8 μm to 2.5 μm 15 μm 200 μm

Near I/R	Mid I/R ^{Fingerprint region} Functional grp region	Far I/R
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Energy $\uparrow\uparrow$ $\downarrow\downarrow$

The mid I/R region (4000 cm^{-1} - 667 cm^{-1}) is used for functional group determination & detect the molecular vibrations

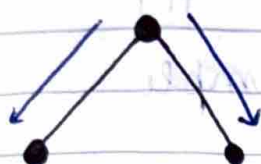
Types of vibrations

Stretching vibration

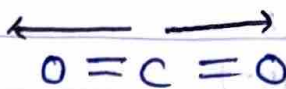
Bending vibrations

→ Stretching vibrations → Req. high energy (1700 cm^{-1})
The atoms move along the bond axis so that the bond length $\uparrow$ or $\downarrow$ periodically.

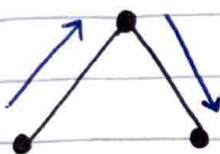
• Symmetric stretching



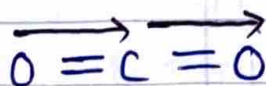
Atoms move in same dirⁿ



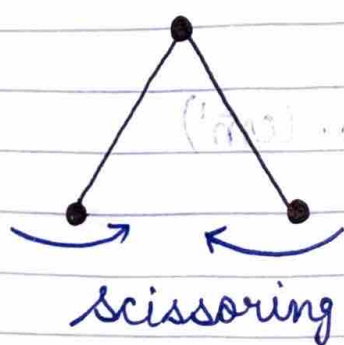
• Asymmetric stretching



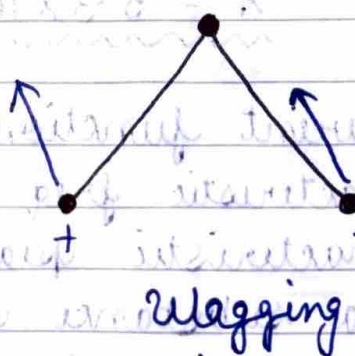
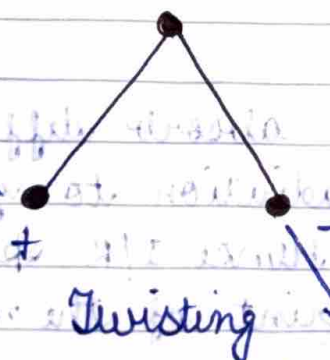
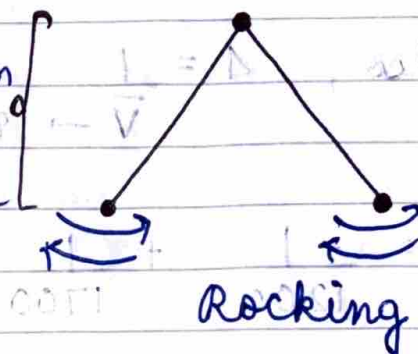
One atom moves towards the central atoms, while other moves away



→ Bending vibrations → Req. low energy (1200 cm^{-1})
The position of the atom changes w.r.t. original bond axis leading to change in bond angle.

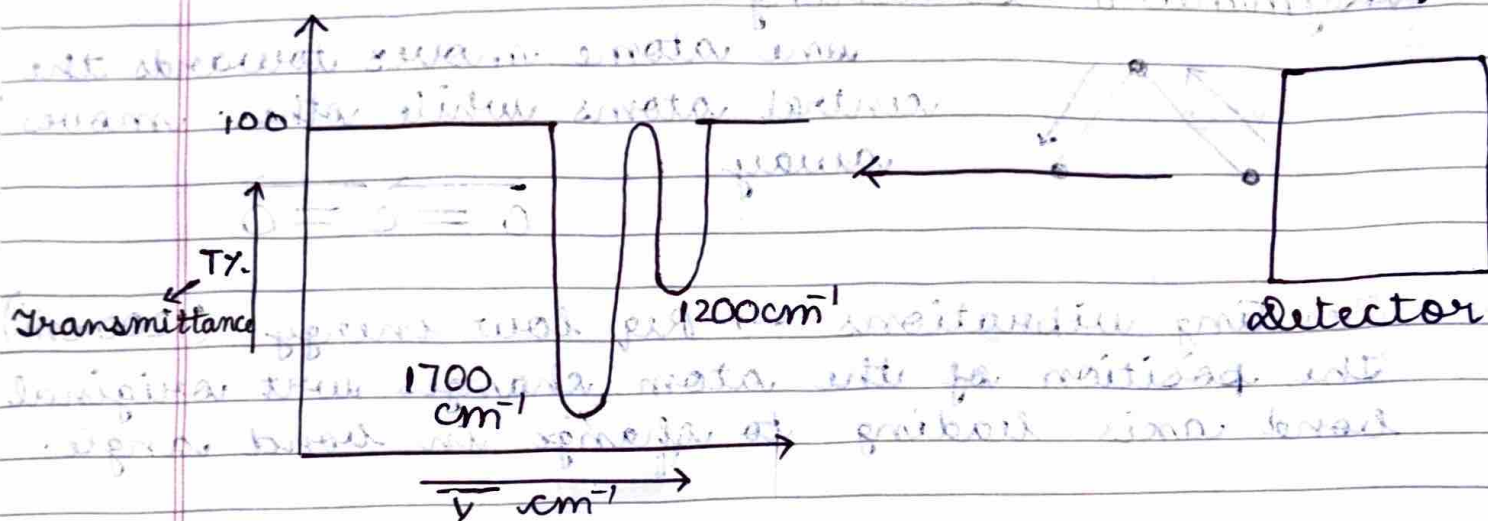
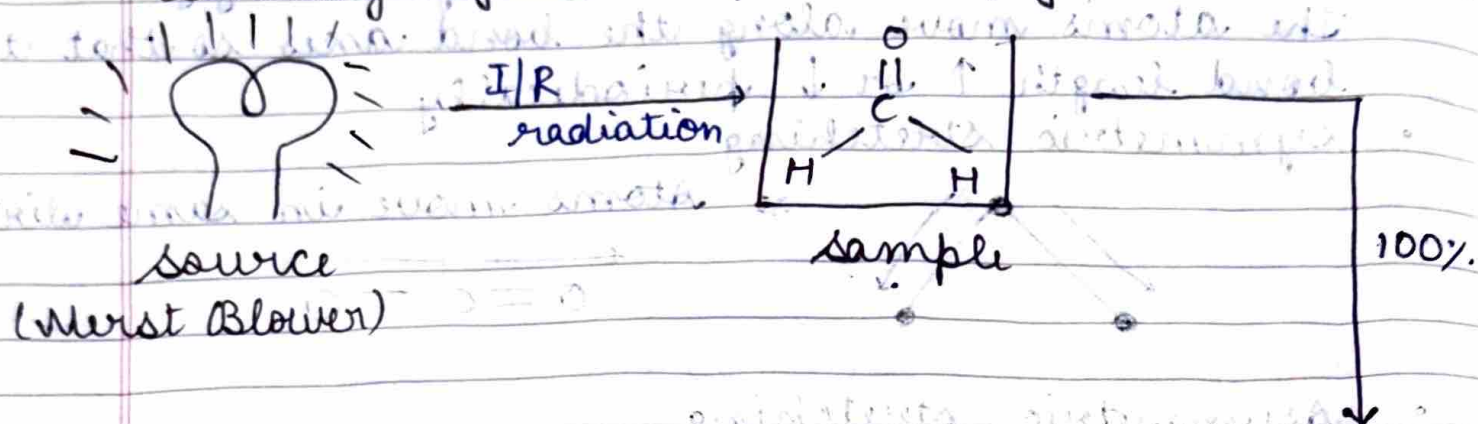


within the plane



out of the plane

→ Working of I/R spectroscopy



Also, $\lambda = \frac{1}{\nu} \rightarrow \text{"wave no. (cm}^{-1}\text{)"}$

$$\Rightarrow \lambda = \frac{1}{1200} + \frac{1}{1700} \text{ cm}^{-1}$$

$$\Rightarrow \lambda = 5.3 \times 10^{-4} \text{ cm}$$

→ Different functional groups absorb different characteristic freq of I/R radiation to give characteristic peak value. Hence I/R spectrum of a substance is fingerprint of the molecule for its determination.

- Applications of I/R spectroscopy
- To find type of group, environment around the group, aromaticity & aliphaticity
 - To find what kind of isomers are present
 - To find molecular structure, chemical kinetics and detection of impurities and conjugation

→ Selection Rule or Transition Rules

Selection rules are used to determine whether a transition is allowed or forbidden.

1. Electronic selection rules

For an atom to absorb or emit a photon then following changes in principal quantum number must occur

- Principle quantum number (n)

$\Delta n = \text{any integer}$, there are no restrictions on changes to energy level.

- Orbital Angular momentum (l) $\Delta l = \pm 1$

This is Laporte's rule. The transitions such as $s \rightarrow p$ and $p \rightarrow d$ are allowed but $s \rightarrow s$ or $s \rightarrow d$ are forbidden.

- Magnetic quantum number (m_l)

$\Delta m_l = 0, \pm 1$

- Spin quantum number (s)

The total spin of system shouldn't change. The singlet to triplet transitions are forbidden.

$\Delta S = 0$

2. vibrational selection Rules (In I/R spectroscopy)

- Gross selection rule

The vibration must cause a permanent dipole

moment of the molecule

- specific selection rule

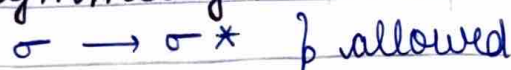
The change in vibrational quantum number must be ± 1

$$\Delta V = \pm 1$$

3. Rotational selection rules (In microwave spectroscopy)
The molecule must have a permanent dipole moment.

$$\Delta J = \pm 1 \rightarrow \text{Rotational angular momentum}$$

4. Symmetry selection rules



Forbidden

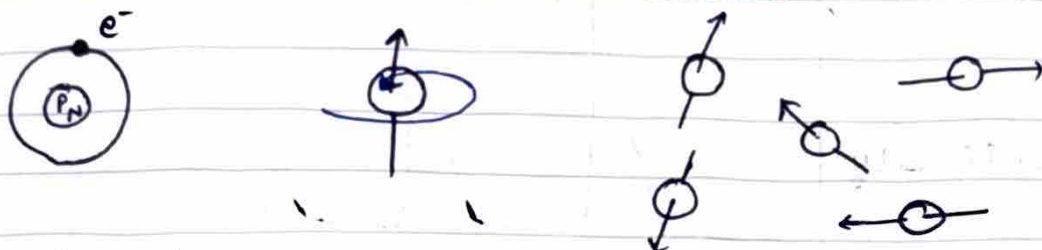
→ Nuclear Magnetic Resonance (NMR spectroscopy)

It's based on the absorption of EM radiations in radio frequency region (4-900 MHz) by nuclei of the atom. It's used to detect the no. of H or protons in compound with accuracy and uses wide variety of nuclei like H, N, C, F, P etc.

→ Elements with either odd mass / odd atomic no. number have property of nuclear spin and are NMR active eg → ^1H , ^{13}C etc

→ Principle of NMR

Spin of a nucleus generates an magnetic field without an external applied field. The nuclear spins are random in direction.



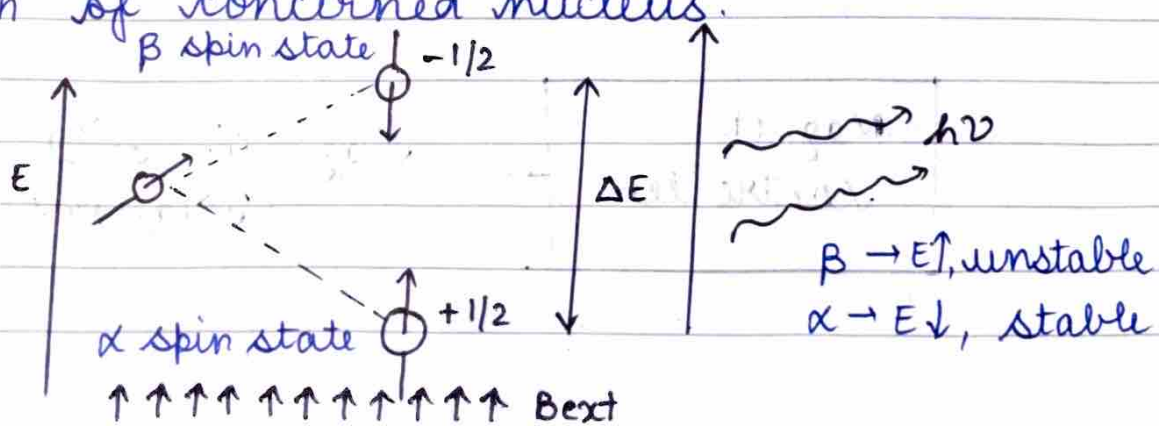
Randomly, creates magnetic fields

• Nuclear spin

Protons have the property of spin like electrons of $+1/2$ or $-1/2$. Both spins have same energy in absence of external magnetic field.

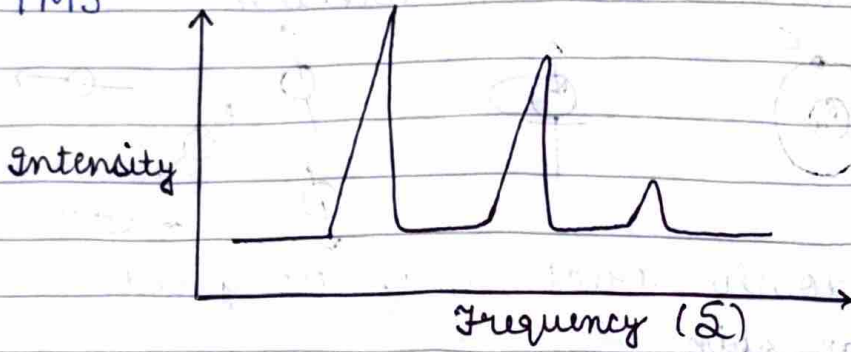
→ When B_{ext} is applied, nuclei align themselves either in same or opposite direction of B_{ext} . When both have same direction then it's called α spin state and it's most stable state.

→ If radio frequency energy is applied, an energy transfer from ground to excited state is possible only when energy of wave matches the energy difference ΔE . This is called resonance when the spin returns to ground state level the absorbed energy is emitted at same frequency and emitted frequency signals gives the NMR spectrum of concerned nucleus.

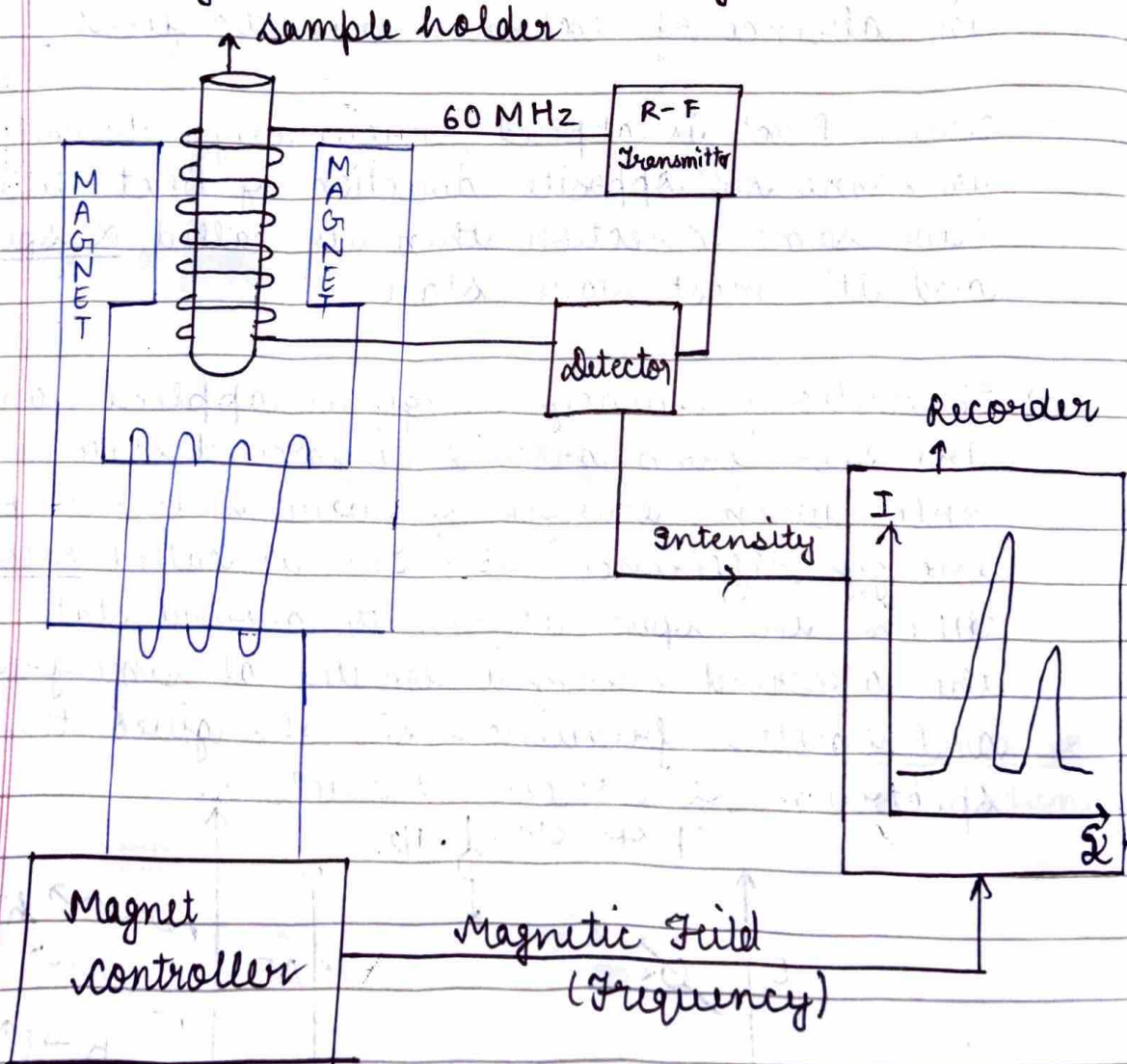


→ NMR spectrum

The NMR spectrum is a plot of intensity of NMR signals versus frequency in reference to TMS



→ working of NMR spectroscopy



→ Sample is placed in a cylindrical glass tube and is dissolved in a proton free solvent like CDCl_3 or CCl_4 and add small amount of TMS for internal reference. This sample is placed between two magnetic poles where coil is attached to a specific R-F generator which supplies radio freq. waves and change orientation of protons. The detector coil detects R-F signals when resonance occurs.

→ As the magnetic field increases linearly, the recorder pen travels from left to right thus protons which achieve resonance faster^(de-shielded), appear on left side (downfield) where those protons achieve resonance slower are shielded and appear on right side (up field) in form of peaks.

→ Use of NMR

All protons show resonance at different frequency because they are surrounded by diff no. of valence electrons which varies with atom so they exist in different environments.

Position of signals help to know nature of proton i.e. aromatic, aliphatic etc. Thus, they absorb at different field strength.

→ Shielding

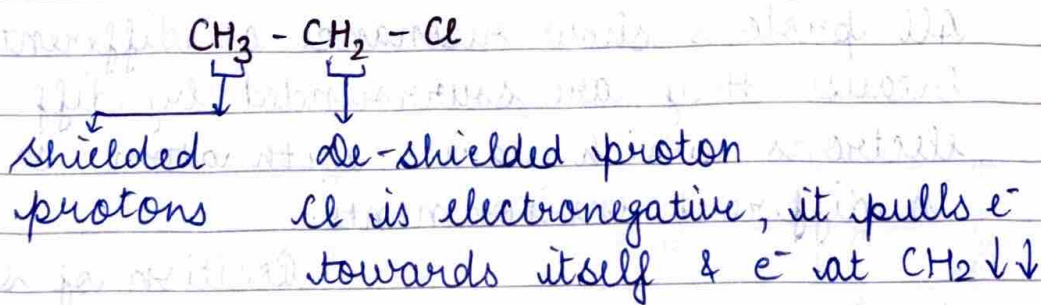
When nucleus is placed in magnetic field so its, surrounding electrons circulate and generate counter field which opposes the applied magnetic field on proton, so field experienced by proton is

reduced and that proton is called a shielded proton. It requires less energy to flip & shifts absorption signal to right side (upfield) of spectrum. This is called as shielding.

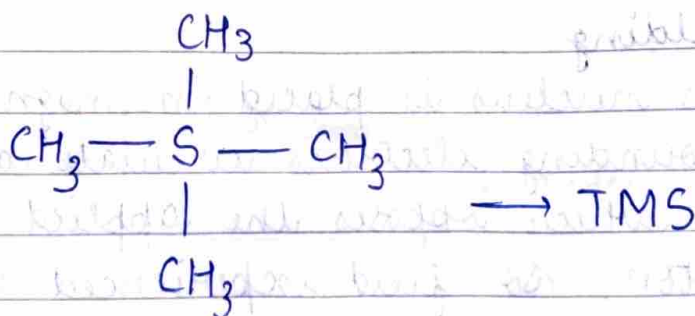
→ De shielding

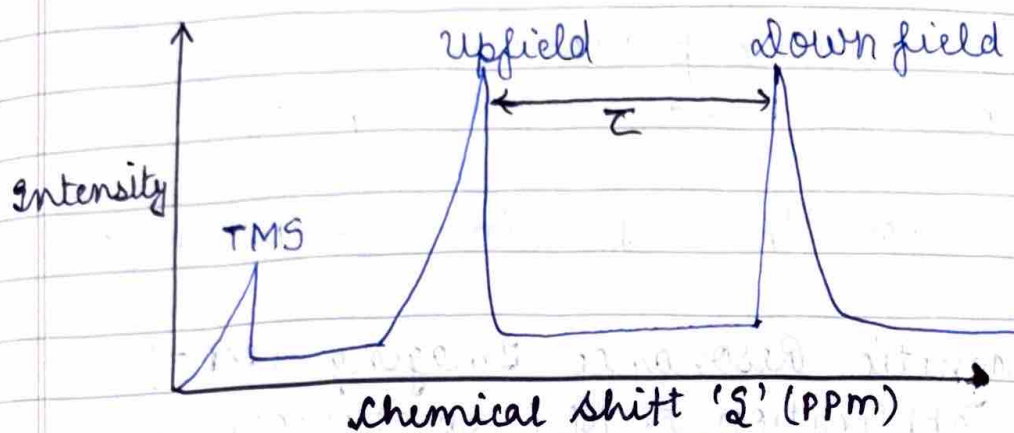
If the e^- density around the nucleus decreases, the opposing magnetic field becomes small and the proton experiences high magnetic field & such proton is called de-shielded proton. It needs high frequency to achieve resonance & signal appears left side (downfield) of spectrum. This is called de-shielding.

→ This process of shielding & de-shielding is called chemical shift. (§)



→ TMS (Tetra methyl silicane) is taken as the reference of chemical shifts because all its protons are similar in nature.





$$\delta = \frac{\text{freq of signal} - \text{freq of TMS}}{\text{spectrometer frequency}} \times 10^6$$

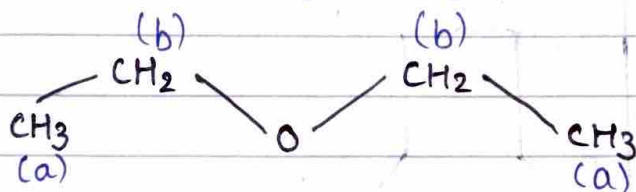
↓
Chemical Shift

→ coupling constant (τ)

Distance between two peaks in a multiplet is a measure of splitting effect known as coupling constant. SI → HZ. It is the measure of effectiveness of spin-spin coupling.

→ spin-spin coupling (splitting)

- Interaction between spins of 2 neighbouring nuclei may cause splitting of NMR spectrum. This is called spin-spin coupling.
- It's related to no. of equivalent H atoms at nearby nuclei; and takes place by $n+1$ rule where $n \rightarrow$ no of neighbouring nuclei



a → shielded, b → de-shielded

For H_a

neighbouring protons = 2

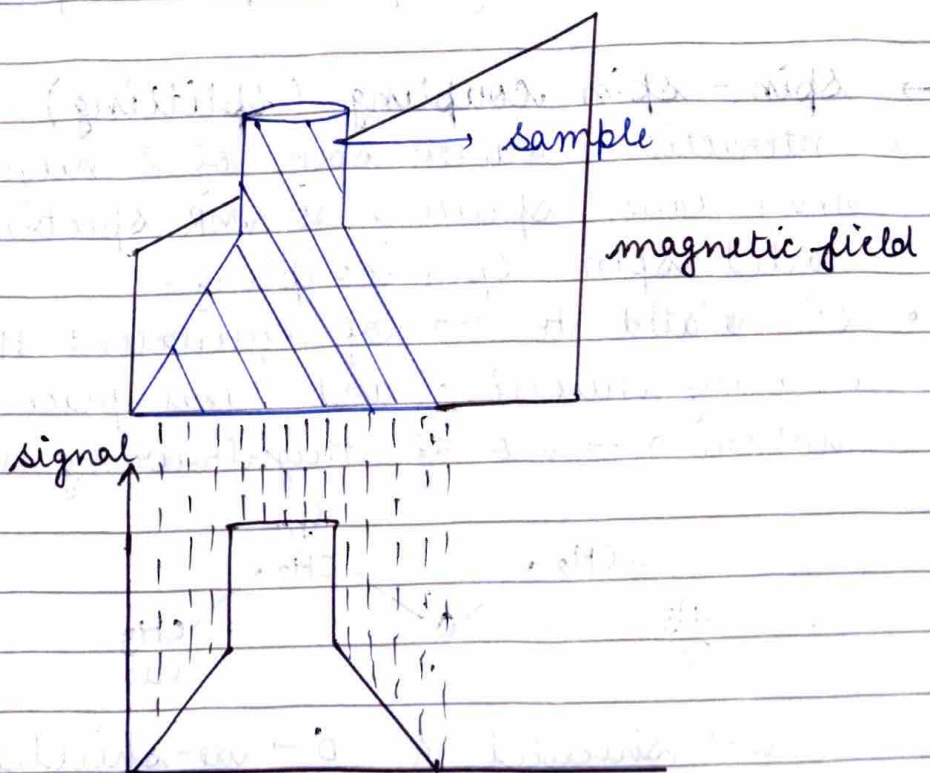
splitting = $2+1=3$ (triplet)

||ly for H_b , splitting = $3+1=4$ (quartet)

→ Magnetic Resonance Imaging (MRI)

It's application of NMR spectroscopy where nuclear magnetic resonance of protons is used to make proton density mapping of body. It helps to differ between healthy & diseased tissues & is based on principle that protons within water lipids resonate at a given frequency. Body has 75% water, with 2 protons for each molecule so images of diff parts can be taken easily.

In diseased tissue, water conc (proton) is $\downarrow\downarrow$ so we can easily detect them.



Structure for MRI

→ Procedure for MRI

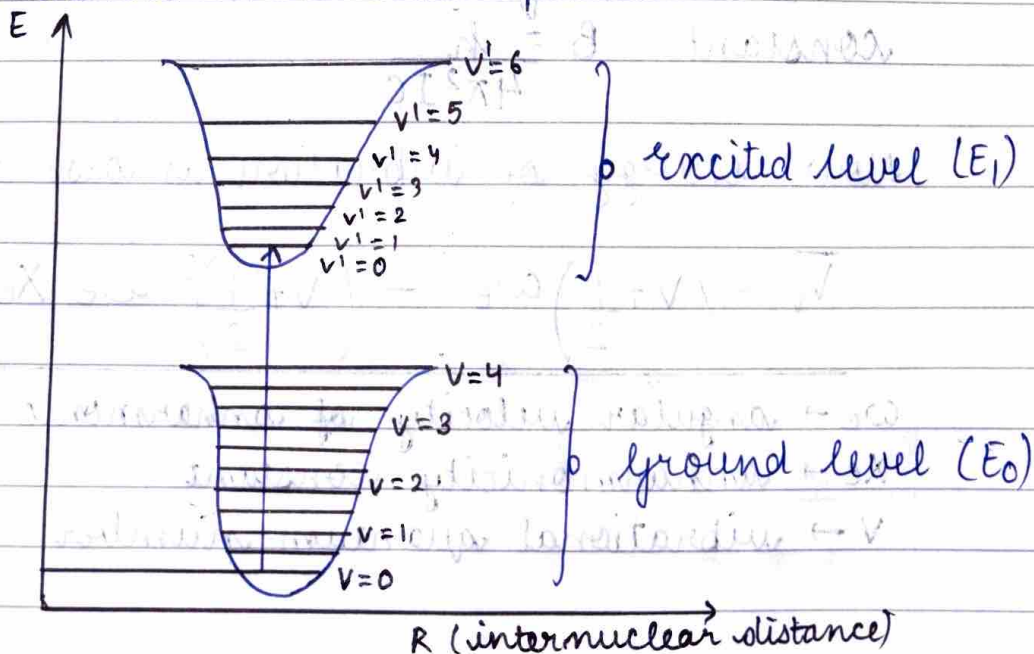
A varying magnetic field is applied across the body part under observation. The proton in diff regions of body attain resonance at different frequencies & intensity of signal is proportional to no of protons at that magnetic field. The body part is taken to different orientation & several projections are taken and combined by computer to get 3D image of body parts in a small time.

→ Applications of MRI

- To investigate functioning of vital organs
- To investigate excessive fat deposition
- For blood analysis & estimation of flourine concentration in different body parts.

→ Franck-Condon principle

It states that the electrons move faster than the nuclei & nucleus is effectively stationary during a transition. This principle is used to represent vibrational-electronic spectra.





→

Rotational-vibrational spectra of diatomic molecules. Radiation energy in I/R region brings about simultaneous change in the rotational and vibrational energies of molecules. This is observed in fine structure of rotational bands and combined spectrum is called vibrational-rotational spectrum.

The rotational & vibrational changes are independent. Thus, net energy is the sum of their energies

$$\underline{\Delta E_{\text{total}} = \Delta E_{\text{vib}} + \Delta E_{\text{rot}}} \quad - (1)$$

Also, energy in terms of wave number,

$$\underline{\bar{\nu}_{J,v} = (\bar{\nu}_J + \bar{\nu}_v) \text{ cm}^{-1}} \quad - (2)$$

Also, energy of rotation is quantized and given by

$$\underline{\bar{\nu}_J = BJ(J+1)} \quad - (3)$$

$J \rightarrow$ rotational quantum number
constant $B = \frac{h}{4\pi^2 I C}$

Now, energy of vibration is also quantized

$$\underline{\bar{\nu}_v = \left(v + \frac{1}{2}\right) \omega_e - \left(v + \frac{1}{2}\right)^2 \omega_e x_e} \quad - (4)$$

$\omega_e \rightarrow$ angular velocity of anharmonic motion
 $x_e \rightarrow$ anharmonicity constant
 $v \rightarrow$ vibrational quantum number

using ③ and ④ in ②

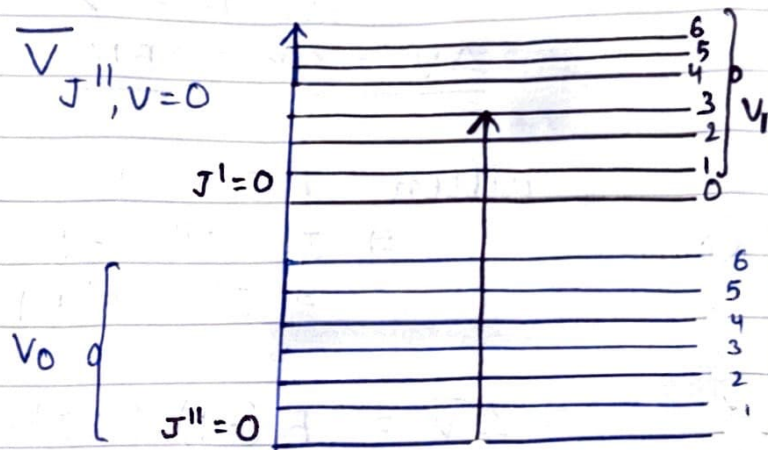
$$\bar{V}_{J,v} = BJ(J+1) + \left(v + \frac{1}{2}\right)\omega_e - \left(v + \frac{1}{2}\right)^2 \omega_e x_e \quad - (5)$$

According to selection rules,

$$\Delta v = \pm 1 \text{ and } \Delta J = \pm 1$$

Now, for vibrational transition from v_0 to v_1 and rotational from J'' to J'

$$\Delta \bar{V} = \bar{V}_{J',v=1} - \bar{V}_{J'',v=0}$$



Now,

$$\bar{V}_{J',v=1} = BJ'(J'+1) + \frac{3}{2}\omega_e - \frac{9}{4}\omega_e x_e$$

and

$$\bar{V}_{J'',v=0} = BJ''(J''+1) + \frac{1}{2}\omega_e - \frac{1}{4}\omega_e x_e$$

$$\text{Now, } \bar{V}_{J',v=1} - \bar{V}_{J'',v=0}$$

$$= \bar{V}_{J',v=1} - \bar{V}_{J'',v=0} = \omega_e - 2\omega_e x_e + B(J' - J'')(J' + J'' + 1)$$

$$\text{Also, now let } \omega_e - 2\omega_e x_e = \omega_0$$

then,

$$= B(J' - J'')(J' + J'' + 1) + \omega_0 = \bar{V}_{J', v=1} - \bar{V}_{J'', v=0}$$

also since $\Delta J = \pm 1$

$$\Rightarrow J' - J'' = 1$$

$$J' = 1 + J''$$

$$\Rightarrow \bar{V}_{J', v=1} - \bar{V}_{J'', v=0} = B(1)(1 + J'' + J'' + 1) + \omega_0$$

$$\Rightarrow B[2J'' + 2] + \omega_0$$

$$\text{let } M = J'' + 1$$

$$\Rightarrow \underline{\Delta \bar{V} = \omega_0 + 2BM} \quad (\text{when } \Delta J = 1)$$

when $\Delta J = -1$

$$\Rightarrow J' - J'' = -1$$

$$\Rightarrow J'' = J' + 1$$

$$\Rightarrow \Delta \bar{V} = B(-1)(2J' + 2) + \omega_0$$

$$= -2B(J' + 1) + \omega_0$$

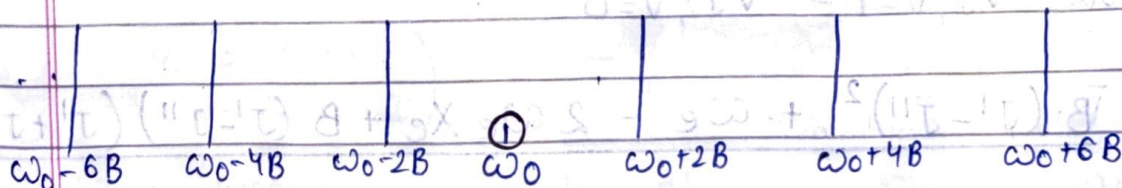
$$\text{let } -(J' + 1) = M$$

$$\Rightarrow \underline{\Delta \bar{V} = \omega_0 + 2BM} \quad (\text{when } \Delta J = -1)$$

where $M = 0, \pm 1, \pm 2$

$$\Rightarrow \Delta \bar{V} = \omega_0, \omega_0 \pm 2B, \omega_0 \pm 4B, \dots$$

Hence, separation is always of $2B$



P branch lines

R branch lines
freq > freq of ω_0

Bragg's Law

$$n\lambda = 2d \sin \theta$$

classmate

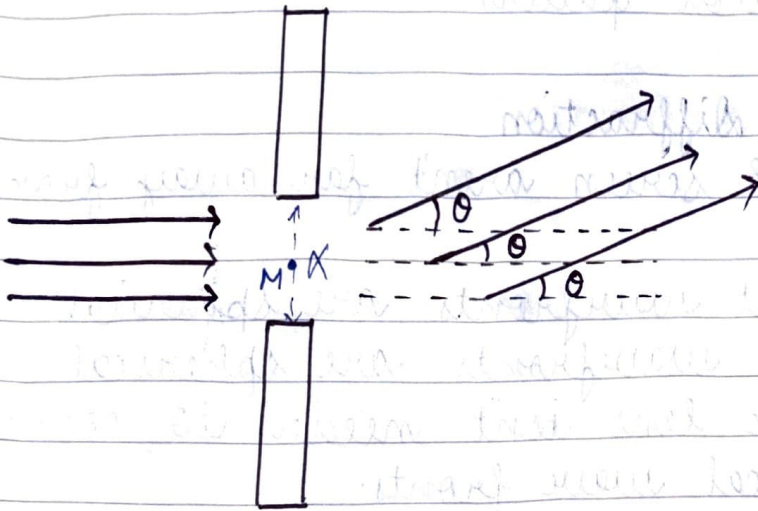
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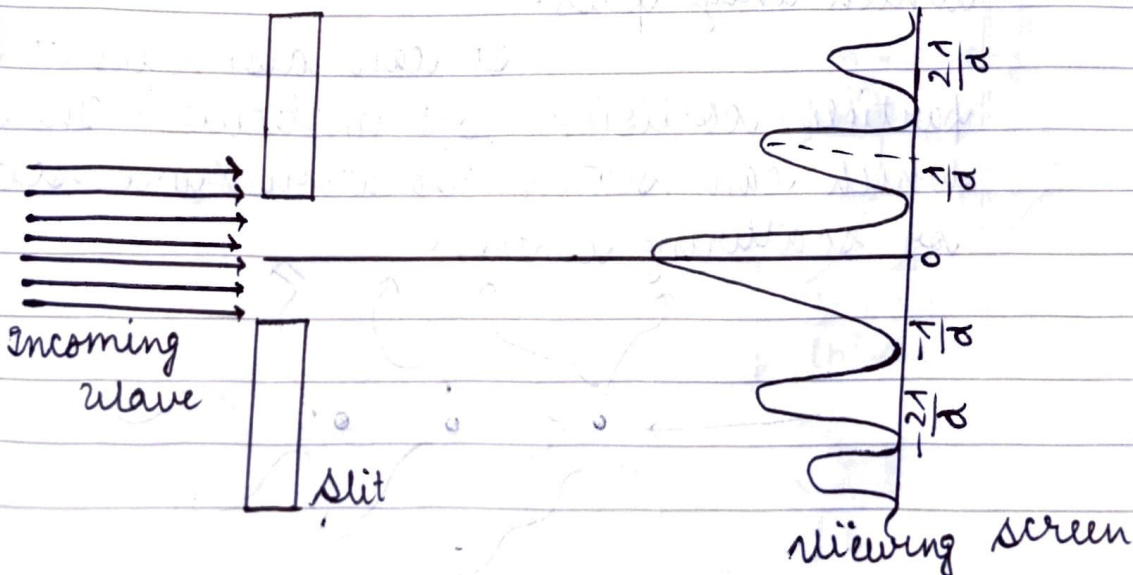
→ Diffraction

spreading out of energy of light wave after it falls on a small slit or hole is called as diffraction.

It takes place only when size of the obstacle is comparable to wavelength of light i.e. from $4 \times 10^{-7} \text{ m}$ to $7 \times 10^{-7} \text{ m}$. If size is near this range, diffraction can be seen.



A parallel beam of light falls normally on slit of width a , diffracted light goes to meet a screen. The mid point of slit is M . A st. line through M $\perp$ to slit plane meets screen at C at central pt. C at screen $\theta = 0^\circ$. All the path diff are zero.



→ Types of Diffraction

1. Fraunhofer Diffraction

- Source & screen are far away from each other
- Incident wavefronts on obstacle are plane
- Output wave fronts after diffraction are also plane
- Plane diffracting wavefronts are converged by means of convex lens to produce a diffraction pattern.

2. Fresnel Diffraction

- Source & screen aren't far away from each other
- Incident wavefronts are spherical
- Output wavefronts are spherical
- Convex lens isn't needed to converge the spherical wave fronts.

→ Scattering

Physical process by which some forms of radiation such as light / sound are forced to deviate from st. line path due to the non-uniformities in medium through which they pass.

It can also refer to particle-particle collisions b/t molecules. The impurities which can scatter radiations are scatterers or scattering centres.

